Signal Dispersion

Concept:

- **Material Dispersion:** This occurs because the refractive index of glass varies with the wavelength of light.
  - Since an LED light source isn't "pure" (it contains a range of wavelengths/colors), different colors travel at different speeds through the fiber.
- **Pulse Broadening:** As these different speeds cause parts of the light pulse to arrive at different times, the pulse "spreads out" or broadens as it travels.
- **Intersymbol Interference (ISI):** For CSE students, the practical impact is **ISI**. If a pulse broadens too much, it starts overlapping with the pulse representing the next bit.
  - The receiver then cannot distinguish between a "1" and a "0".

Exercise: Material Dispersion and Pulse Broadening

Question:

A glass fiber has a material dispersion coefficient (D_{mat}) of **80 ps/(nm·km)** at an operating wavelength of **850 nm**. The light source used is an LED with a spectral width ($\Delta\lambda$) of **40 nm**.

1. Calculate the **pulse broadening (ΔT)** due to material dispersion for a fiber length of **2 km**. (2 Marks)
2. Calculate the maximum **Bit Rate (R_B)** that can be supported by this link, assuming the bit rate is limited to $R_B \approx 1/(2\Delta T)$. (2 Marks)

Solution

1. Calculating Pulse Broadening (ΔT):

The formula for pulse broadening due to material dispersion is:

$$\Delta T = D_{mat} \times L \times \Delta\lambda$$

Where:

- D_{mat} = $80\ \mathrm{ps/(nm{\cdot}km)}$
- L (Length) = $2\ \mathrm{km}$
- $\Delta\lambda$ (Spectral Width) = $40\ \mathrm{nm}$

$$\Delta T = 80 \times 2 \times 40$$

$$\Delta T = 6400\ \mathrm{ps} = \mathbf{6.4}\ \mathrm{ns}$$

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2. Calculating Maximum Bit Rate (R_B):

Using the provided approximation for a dispersion-limited link:

$$R_B = \frac{1}{2\Delta T}$$

$$R_B = \frac{1}{2 \times 6.4 \times 10^{-9}\ \mathrm{s}}$$

$$R_B \approx 78,125,000\ \mathrm{bps}$$

$$R_B \approx \mathbf{78.13}\ \mathrm{Mbps}$$